

BLOCK 7

TRADE, GROWTH AND DEVELOPMENT

Unit 16	Debate on the Trade and Growth Nexus
Unit 17	Trade and Environment

BLOCK INTRODUCTION

This block is on **Trade, Growth and Development**. It has two units, i.e., Unit 16 - **Trade and Economic Growth Nexus** and Unit 17 – **Trade and Environment**.

Unit 16 provides us an understanding of the link between the economic growth and trade. In order to do so, the unit starts with the discussion on the importance of economic growth with reference to the theoretical underpinnings in terms of classical and endogenous models, trade and Solow model, trade and productivity growth. The unit moves in the direction of explaining various trade policy regimes vis-à-vis growth in developing countries wherein the import substitution and outward oriented development strategies are discussed. At the end of the unit, India's experience with the inward-oriented and outward-oriented regime is presented.

Unit 17 takes the theme ahead by providing us the link between trade and environment wherein the role of externalities has been outlined. The unit also discusses the link of trade with climate change and the policies and practices in that regards. The relevance of WTO in safeguarding the environment has been highlighted and, in the end, the multilateral environmental agreements in relation to trade are discussed.

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UNIT 16 DEBATE ON THE TRADE AND GROWTH NEXUS *

Structure

- 16.0 Objectives
- 16.1 Introduction
- 16.2 Importance of Economic Growth
- 16.3 Sources of Economic Growth: Theoretical Underpinnings
 - 16.3.1 Theories of Growth: Classical and Endogenous Growth Models
 - 16.3.2 Trade and Growth in the Solow Model
 - 16.3.3 Trade and Productivity Growth: Theoretical Links
- 16.4 Trade Policy Regime and Growth in Developing Economies
 - 16.4.1 Criticism of Primary-Export-Led Growth Policies
 - 16.4.2 Import Substitution Industrialisation Strategy
 - 16.4.2.1 Country Experiences
 - 16.4.2.2 Criticism of ISI
 - 16.4.3 Outward Oriented Development Strategies
 - 16.4.3.1 Country Experiences
 - 16.4.3.2 Critical Evaluation
- 16.5 Indian Experience
 - 16.5.1 Experience with the Inward-Oriented Industrialisation Regime
 - 16.5.2 Experience with the Outward Oriented Liberalisation Regime
- 16.6 Let Us Sum Up
- 16.7 Key Words
- 16.8 Some Useful References
- 16.9 Answers/Hints to Check Your Progress Exercises

16.0 OBJECTIVES

After studying this Unit, you should be able to:

- Explore the relevance, nature and determinant of economic growth in general;
- Examine the relationship between international trade and economic growth in classical and endogenous growth theories;
- Identify the link between international trade and productivity growth;

* Contributed by Dr. Rijesh Raju, Institute for Studies in Industrial Development (ISID).

- Discuss the issue of trade and growth nexus in the light of trade policy regime in developing countries;
- Highlight the transition and country-level experiences of import substitution and outward-oriented policy regimes; and
- Provide an overview of the trade policy regime and growth experience of Indian economy since economic planning.

16.1 INTRODUCTION

Economic growth is of fundamental significance for any economy. An economy is said to grow when the amount of goods and services produced in the country expands over time. The total amount of output produced for final consumption in a particular period, normally a year or the economy's income level is referred to as Gross Domestic Product (GDP). Since every economy experiences growth in labour force/population, improving the rate of growth of GDP matters most for achieving sustained higher standards of living in the long run. In this sense, GDP is measured by the rate of change of real income per capita or simply, output per labour input (i.e., labour productivity). In this unit, we will learn how growth rate of an economy is related with its trade volume. We will try to understand how it affects each other.

16.2 IMPORTANCE OF ECONOMIC GROWTH

Economists argue that growth in GDP enables an economy to improve the overall welfare of the people and economic development. We know that the welfare of the citizen ultimately depends on improvement in the quality of life, including higher levels of consumption, better educational achievements, quality housing and better access to valuable health facilities. Since a sustained level of economic growth is highly correlated with higher standard of living and economic welfare, GDP growth is often considered as a key instrument (or means) for achieving economic development (the end goal).

Rather than the level of GDP, it is the rate of increase in GDP which matters most. This is because of the well-established evidence that points out that prolonged differences in GDP growth rates result in dramatic changes in living standards. One useful summary measure to appreciate the significance of rate of GDP growth is to find out the number of years it will take the size of the economy to double over time. According to a simple approximation of time, the number of years it takes for an economy to double its level is equal to 70 divided by the growth rate. This is known as "*Rule of 70*". According to this rule, an economy with a GDP growth rate of 1 per cent doubles its living standard (i.e., doubles the economy's overall size) every seventy years, while an economy with a rate of growth of 5 per cent per annum doubles its size and living standard every fourteen years.

Examples of such drastic differences in growth rates among countries are numerous. For example, in the mid-1950s, the per capita GDP of El Salvador and Japan were roughly similar. However, by 1993, the income of one Japanese person was approximately equal to that of 24 Salvadorans, which was achieved through a sustained level of per capita growth rate of less than 9 per cent per annum maintained over 38 years by the Japanese economy. This example illustrates that a percentage point drop in the growth rate makes a huge difference in the long period. Therefore, it is important to appreciate and understand the factors determining the rate of growth in GDP in an economy. In the following Section, we will attempt to provide an answer based on the standard theories of economic growth.

16.3 SOURCES OF ECONOMIC GROWTH: THEORETICAL UNDERPINNINGS

16.3.1 Theories of Growth: Classical and Endogenous Growth Models

Identifying the sources of income generation is one of the main focuses of the growth and development theories. An early attempt to understand the principal source of economic growth is attributed to the work of Prof. Robert Solow¹ in the early 1950s. The *Classical Growth Theory*, based on the work of Solow and many others, have tried to disentangle the contribution of growth in physical capital stock, growth in the labour supply and state of technology in generating income per capita over time.

Using the standard production function approach, the classical theories argue that in the long period, the rate of savings determines the size of capital stock and thus the level of output. The model predicts that an increase in saving rates will have a corresponding *level effect* on income per person (labour productivity) as output expands, but will eventually reach a steady state² equilibrium level. Alternatively, an increase in labour supply or population growth will negatively impact the steady-state levels of both capital per worker and output per worker. The Solow model predicts that regardless of the availability of factor inputs, the economy will tend to move towards a steady state long run equilibrium level. Since the level of savings determines the stock of capital, a large supply of savings allows the accumulation of productive capital and a higher output level in the steady state. However, once the economy reaches the steady state, policy

¹The Solow growth model is named after Robert M. Solow on his work on the accounting the sources of economic growth in the 1950s. The model was introduced in the paper Solow, R. M. (1956). A Contribution to the Theory of Economic Growth. *The Quarterly Journal of Economics*, 70(1), 65-94. In 1987, Solow won the Nobel Prize in economics for his work on economic growth.

² Steady state is a condition in which the key economic variables are constant and does not deviate.

instruments are needed to alter the steady-state growth rate of income per person to have a *growth-effect* over time. In effect, the policy option is to choose the steady state that provides maximum welfare to the society in terms of the availability of goods and services for consumption. The steady-state level of capital stock that maximises the level of consumption is called *Golden Rule Level of Capital*. The classical models show that if the level of capital stock is larger than the Golden rule steady state, then policies that reduce saving will increase consumption at all time periods.

However, since the capital stock is subject to decreasing returns, the capital accumulation itself cannot sustain the growth rate, as a continuous rise in output per worker requires a far greater increase in capital per worker. Naturally, over time, the economy may be subject to saving rate constraint and limit further accumulation of productive capital. In this case, for sustainable economic growth, it is necessary to have a continuous progress in the state of art of technology in production. Accordingly, the classical theories build models including exogenous technological progress, which over time expands the society's production capabilities.

A fundamental way to incorporate exogenous technological change is to assume that technology is embodied in workers skill that increases the efficiency of labour. As society's knowledge pool increases, the efficiency of labour increases and the labour per hour output of goods and services expands considerably. The exogenous rise in labour efficiency can result from new assembly line production methods, labour training, health, education skills, etc. In this *labour augmenting technological progress* model, the technological change causes the effective number of productive labours to increase over time. Therefore, at the steady state level of capital per effective labour, the rates of actual output per labour increases at the rate of labour augmenting technical progress. Thus, according to the Solow model, once the economy reaches the steady state equilibrium, the rate of growth of output per worker depends only on the rate of technological progress. In the long run, *the level of technological progress* explains sustained growth and persistent rise of standard of living in an economy.

The classical growth theories show clearly that the sustained output growth per worker requires continuous progress in the state of art of technology. The Solow model, however, considered technical change as exogenous; the model does not explain the determinant of technology. As such, the model often failed to account for many empirical realities such as cross-country income convergence, international migration pattern and large flow of capital across countries. The limitation of the Solow model in explaining the growth experiences of both the developed and the developing world prompted economists to formulate models that treat as endogenous the generation and diffusion of technology. The models that reject the over simplifying assumption of exogenous technology and explain

the determinant of technological advances is known as the *Endogenous Growth Theory*.

The endogenous theoretical models were developed since the 1980s by the scholarly writings of Paul M. Romer, Robert E. Lucas, Jr, among others.³ Instead of assuming diminishing returns for physical capital stock, the basic endogenous model incorporates knowledge as a form of capital with constant returns to scale. These models try to explain the decisions that determine knowledge creation through Research and Development (R&D). A main feature of the endogenous growth model is that they treat research as an economic activity that requires productive input resources that respond to profit opportunities. In the earlier models of Romer and others, technological progress arises from purposeful investment by the profit-seeking firms in R&D and externalities (spill over effects) from such activities. Since technological knowledge is a non-rival and partially excludable commodity, the spill over channel plays an important role in determining the rate of growth of economies over time.

16.3.2 Trade and Growth in the Solow Model

We can use the Solow model to depict the effect of international trade on economic growth. Here we will follow the model proposed by Baldwin (1992)⁴. Under the standard models of international trade, a shift from autarky to free trade increases the real income and welfare in an economy. Assuming the country produces two commodities, the Heckscher-Ohlin (H-O) model of trade shows that opening up of trade increases the real income. This is depicted by the production Possibility Frontier (PPF) movement where the nation specialises and reallocates resources in commodities that have a relative (comparative) cost advantage, determined by factor endowments. The specialisation pattern also allows consumer welfare to rise because compared to the situation in autarky, the consumption points of both commodities in free trade lie under a higher indifference curve.

According to Baldwin (1992), a one-time improvement in welfare from autarky to free trade generates a secondary improvement in real output. The secondary improvement results from increased productive efficiency that transforms available (given) inputs into welfare-enhancing final products. This productivity improvement shifts the PPF in the Solow model and generates economic growth as economy shifts up to a higher steady state equilibrium level of capital and output.

³ See Romer, P. M. (1986). Increasing Returns and Long-Run Growth. *Journal of Political Economy*, 94(5), 1002-1037; and Lucas, R. E. (1988). On the Mechanics of Economic Development. *Journal of Monetary Economics*, 22(1), 3-42.

⁴The full reference for the paper is: Baldwin, R. E. (1992). Measurable Dynamic Gains from Trade,. *Journal of Political Economy*, 100(1), 162-174.

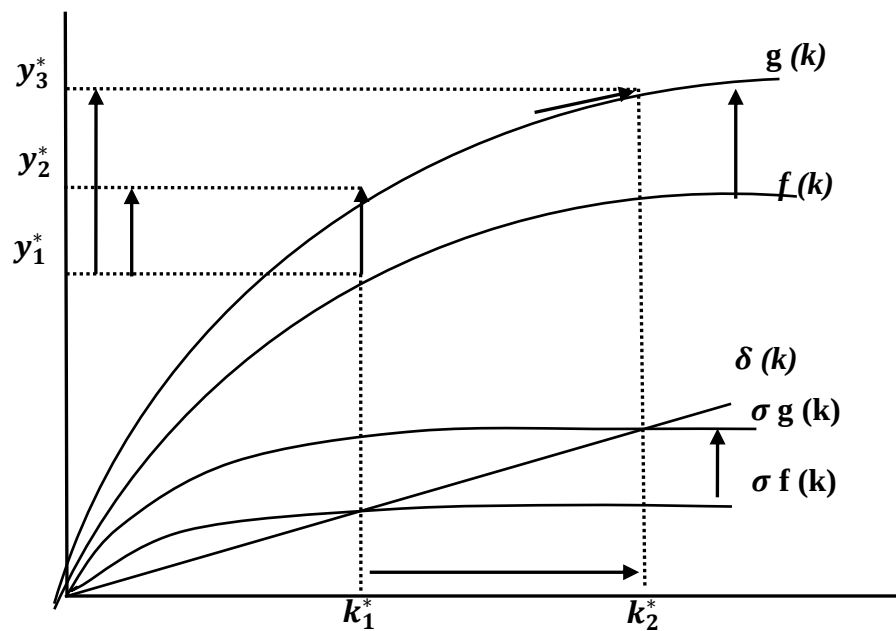


Figure 16.1 International Trade and Solow Growth Model

Source: Adapted from Berg and Lewer (2007)

The process of trade under the Solow growth model is represented in Figure 16.1. The international exchange of goods induces the PPF curve to shift from $f(k)$ to $g(k)$. This changes the economy's steady state capital-output ratio and per capita income from k_1^* and y_1^* to k_2^* and y_2^* , respectively. These changes are the result of an immediate increase in per worker output from y_1^* to y_2^* before the changes in capital-output ratio. The change in y_1^* to y_2^* is the *Static Gains from participating in International Trade*. However, the economy may further increase the gains from trade provided an increase in real income is associated with a rise in economy's saving functions. In this case, the steady state equilibrium shifts upward and generates a secondary spurt of medium-run growth that allows the economy to move to the new steady state equilibrium at k_2^* and y_3^* . The medium-run spurt may be influenced by the composition of trade, namely, whether a country's comparative advantage belongs to the production of capital goods or consumer goods. However, this model shows that the traditional gains from trade is not permanent as the economy eventually reaches and settles at the new steady state equilibrium.

However, instead of the *level effects*, the new growth theory and theoretical advancements in trade provides several plausible channels through which international trade can induce sustained *growth effects*. Based on the theoretical literature, we provide a brief overview of the plausible channels through which trade impacts productivity in the following sub-section.

16.3.3 Trade and Productivity Growth: Theoretical Links

The theoretical literature has identified several channels through which international trade impacts domestic productivity growth⁵. International trade allows domestic producers to achieve *economies of scale effects* relative to autarky. Economies of scale are realised as production moves at a lower average unit cost curve through the reduction of average fixed cost per unit of output and a downward shift of cost curve resulting from additional fixed investments made to meet the additional expectation of market opportunities abroad. Compared to autarky, a liberal trade regime induces *market competition effects* that boost firm efficiency. The entry of foreign competitive products acts as a market disciplining effect on import-competing domestic firms. For market survival, firms will be forced to invest in cost reducing activities such as technology upgradation, skill development and reduce managerial slackness. The exit of inefficient firm and reduction in monopoly power will eliminate X-inefficiency. The lower priced imports improve productivity through investment in R&D, corporate restructuring, and learning from foreign products.

Trade allows countries to specialise and follow their true comparative advantage. The traditional trade theory envisages inter-industry specialisation, whereas recent theories predict large scale within sector production reallocation from less efficient and more productive firms. The *resource reallocation effects*, induced by plant-level heterogeneity, will provide substantial dynamic productivity benefits. Since exporting involves heavy sunk-cost, it is argued that only productive firms will enter in the export market. As exporting firms share increases, the average industry productivity improves. Finally, international trade acts as a vehicle of transmission of knowledge to the recipient economy. As we noted in the earlier section, trade provides *technology spill over effects* that augment aggregate knowledge stock.

So far, we have discussed the theoretical channels through which international trade augments economic growth in an economy. In practice, countries use different policy instruments to either participate or restrict the flow of goods and services across borders. As such, the nature of trade policy initiatives adopted by the government has important implications on the pattern of trade and growth for the development of catch-up developing economies.

⁵For a detailed discussion and literature survey, see the following articles (a) Rijesh, R. (2017) International Trade and Productivity Growth: Evidence from the Organised Manufacturing Sector in India', ISID Working Paper 198, Institute for Studies in Industrial Development, New Delhi; and (b) Singh, T. (2010). Does International Trade Cause Economic Growth? A Survey. *The World Economy*, 33(11), 1517-1564.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Why is it important to understand the factors determining the rate of growth in GDP in an economy?

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- 2) How can we use the Solow model to depict the effect of international trade on economic growth?

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- 3) Highlight the theoretical links between trade and productivity growth.

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16.4 TRADE POLICY REGIME AND GROWTH IN DEVELOPING ECONOMIES

The successful industrialisation of the western economies has forced the policy makers in the less developed economies to plan appropriate strategy for catch-up and development. Since, by definition, the less income countries have low purchasing power, the industrial development process is naturally constrained by insufficient market demand. The market constraints will lead to underutilisation of economic resources, especially in which the country has comparative advantage, especially from the availability of fertile land and resources. We know that most of the low-income countries have abundant natural resources that are ideal for producing resource-or labour-intensive products. Since increasing market access allows countries to specialise according to comparative advantage, producing and exporting labour-intensive products involves a **Primary-Export-Led Development Strategy**. This involves the support of the government to design policies that favour exploitation of natural comparative advantage in exchange of manufactured goods produced in the rest of the world.

The primary export-led growth strategy based on a country's natural economic resources is perceived to improve economic welfare as the production sector specialises and resources are reallocated according to the static gains from trade.